

Guilherme S. G. Brotto

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Professional Objective

Computer Science graduate with a strong focus on Artificial Intelligence and Deep Learning. Experienced in developing machine learning systems, generative models, and data-driven applications. Interested in applying modern AI techniques to real-world problems involving data analysis, automation, and network traffic monitoring.

Education

Federal University of Espírito Santo (UFES) – Vitória, Brazil

B.S., Computer Science: June 2021 - March 2026

Università di Torino – Turin, Italy

Visiting Undergraduate Research Student: April 2025 – July 2025

Research internship abroad funded by the São Paulo Research Foundation (FAPESP).

Professional and Research Experience

Automation Assistant – Mindworks: Oct 2025 – Present

- Develop automation systems for corporate processes using n8n and database technologies.
- Design and implement automated workflows integrating multiple services and APIs.
- Optimize operational tasks through logic-based automation and data-driven solutions.
- Build integrations between internal systems and external platforms to streamline business processes.

Software Developer – Nicchio Café S.A. Exportação e Importação

Aug 2024 – Apr 2025

- Designed and implemented a web server to manage sales operations and currency exchange contracts.
- Developed internal tools to automate business workflows and manage financial data.
- Technologies: Python (Flask), SQL, JavaScript, LDAP.

Scientific Initiation Researcher – Laboratório de Pesquisa em Redes e Multimídia (LPRM), UFES

Nov 2023 – Feb 2026

- Researched generative models for synthetic network traffic and IoT monitoring data.
- Implemented and evaluated Generative Adversarial Networks (GANs) for time-series data generation.
- Developed evaluation metrics to measure fidelity, utility, and statistical similarity of generated data.
- Conducted experimental comparisons across multiple generative models.
- Contributed to peer-reviewed publications in international conferences.

Teaching Assistant – Advanced Structured Programming (UFES)

2022 – 2023

- Assisted in the Programming II course focused on the C programming language.
- Graded programming assignments and projects.
- Supported students with debugging, algorithms, and programming concepts.

Relevant Projects

FlowMIA: Evaluating Membership Inference Attacks on Generative Models

Undergraduate thesis project evaluating privacy leakage in tabular generative models using Membership Inference Attacks.

Episodic Memory for Generative Pre-Trained Transformers

Implemented an episodic memory mechanism for GPT-based models using PyTorch and Sentence-Transformers.

AI Chrome Dino Game

Developed an intelligent agent to play the Google Chrome Dino game using MLP and Genetic Algorithms.

Hackathon SBRC 2024 – 2nd Place

Blockchain-based decentralized marketplace for solar energy credit trading using Cartesi Rollups.

Brain Tumor Segmentation in MRI Scans

Neural network for classification and segmentation of tumors in MRI images using CNNs and PyTorch.

Technical Skills

Programming Languages: Python, C, C++, Java

AI / Machine Learning: PyTorch, Scikit-learn

Libraries & Tools: NumPy, Pandas, OpenCV, Matplotlib, Jupyter Notebook

Databases: MySQL, SQL Server

Key Concepts: Deep Learning, Generative Models, Time Series Analysis, GANs, CNNs, RNNs, LSTM, Transformers

Publications and Conferences

Geração de Dados de Ataque em Internet das Coisas utilizando Redes Generativas Adversárias – VIII Workshop de Computação Urbana (2024)

IoTTraffic: Assessing Generative Models for Internet of Things Attack Data Flows – IEEE WF-IoT (2024)

Measuring Fidelity and Utility of Time Series Generative Adversarial Networks – IEEE ISCC (2024)

Certifications

Sustainable Development Goals in Supply Chain Management – UFES & Texas A&M University (40h)

SBRC 2024 Hackathon – 2nd Place – Sociedade Brasileira de Computação (SBC)

Languages

Portuguese — Native

English — Fluent

Italian — Intermediate

Spanish — Basic